MongoDB, Dockerfile & Java Program

1. MongoDB Commands

```
// 1. Show databases
show dbs

// 2. Insert a student
db.student.insertMany([
  { name: "Naveen", age: 19, city: "Ballari" }
])

// 3. Display all students
db.student.find()

// 4. Insert another student
db.student.insertMany([
  { name: "Naveen", age: 19, city: "SGP" }
])

// 5. Find one student
db.student.findOne({ name: "Naveen" })

// 6. Update student's age
db.student.updateOne(
  { name: "Naveen" },
  { $set: { age: 22 } }
)

// 7. Delete one student
db.student.deleteOne({ name: "Naveen" })

// 8. Find Naveen
db.student.findOne({ name: "Naveen" })

// 9. Insert one student
db.student.insertOne({
  name: "Ravi",
  age: 20,
  city: "Bangalore"
})

// 10. Find students from SGP
db.student.find({ city: "SGP" })

// 11. Update Naveen's city
db.student.updateOne(
  { name: "Naveen" },
  { $set: { city: "Bangalore" } }
)
```

2. Dockerfile

```
FROM eclipse-temurin:25-jdk
WORKDIR /app
COPY Sample.java .
RUN javac Sample.java
CMD ["java", "Sample"]
```

3. Sample.java

```
public class Sample {
    public static void main(String[] args) {
        System.out.println("Hello, World!");
    }
}
```

4. Docker Commands

```
# Build the Docker image (note the space before the final dot)
docker build -t myjavaapp .

# List Docker images
docker images

# Run the Java application
docker run --rm myjavaapp
```

Note: Save the Java source file as Sample.java because the public class is named Sample. Run Docker commands from the folder containing both Sample.java and Dockerfile.